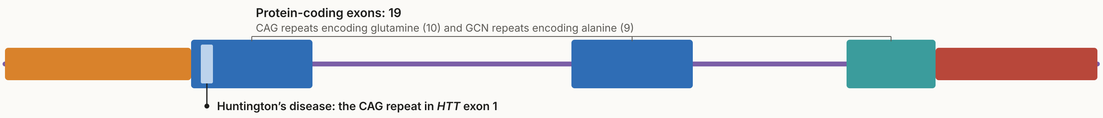
Repeat expansion diseases: normal and disease-causing repeat lengths

Lengths a person is born with, in repeat units, on a logarithmic scale. For Huntington's disease the disease-causing range starts at 36, a threshold known since the gene was found in 1993. Forty or more always leads to the disease; 36 to 39 carries a reduced risk that rises with age, and some carriers never develop it. The dashed line at 150 is a different measurement, made on a single neuron decades later: by then the repeat in that cell has grown well beyond what the person inherited, and cells in one brain differ widely. Above about 150 in a given cell, that cell begins to change. Only Huntington's disease has been measured this way.



Promoter and 5′ untranslated region: 13
CGG, CCG, CAG, GCA, GGGGCC, 12-base unit

Introns: 15 (3 without reported lengths)
GAA, CCTG, ATTCT, ATTTC, AAGGG, GGCCTG, CTG, CCCTCT

3′ untranslated region: 2
CTG (myotonic dystrophy 1), CAG/CTG (SCA8)

Plus two longer repeat units (25 and 99 bases) in an exon and an intron. Counts are the disorders listed by Depienne and Mandel.
The table below draws 47 of them, merging the two fragile X premutation disorders into one row and omitting three without published ranges.

Normal range · Disease-causing range, coloured by where the repeat sits in the gene · Only a minimum reported

150 CAGs: what the repeat has grown to in one neuron late in life, not a length anyone is born with

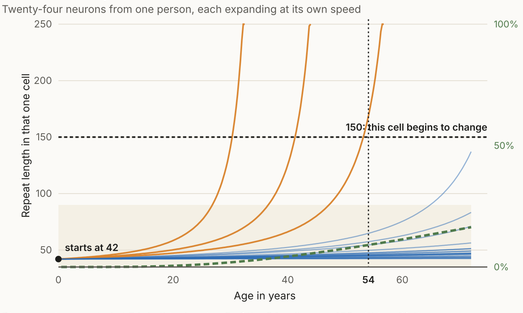
Disease	Gene	Normal → disease
Protein-coding sequence: CAG repeats (glutamine tracts)		
Spinal and bulbar muscular atrophy	*AR*	9–36 → 38–68
Huntington's disease	*HTT*	6–35 → 36–250
Spinocerebellar ataxia 1	*ATXN1*	6–38 → 39–88
Dentatorubral-pallidoluysian atrophy	*ATN1*	3–35 → 48–93
Spinocerebellar ataxia 2	*ATXN2*	13–31 → 32–500
Spinocerebellar ataxia 3 (Machado-Joseph)	*ATXN3*	12–44 → 55–87
Spinocerebellar ataxia 6	*CACNA1A*	4–18 → 20–33
Spinocerebellar ataxia 7	*ATXN7*	4–33 → 37–460
Spinocerebellar ataxia 17	*TBP*	25–40 → 43–66
Huntington's disease-like 2	*JPH3*	6–28 → 41–58
Protein-coding sequence: GCN repeats (alanine tracts)		
Synpolydactyly 1	*HOXD13*	15 → 24
Brachydactyly and cleidocranial dysplasia	*RUNX2*	17 → 27
Oculopharyngeal muscular dystrophy	*PABPN1*	6–10 → 12–17
Hand-foot-genital syndrome	*HOXA13*	18 → 24–30
Blepharophimosis, ptosis, epicanthus inversus	*FOXL2*	14 → 19–24
Holoprosencephaly 5	*ZIC2*	15 → 25
Infantile epileptic encephalopathy 1	*ARX*	16 → 23
Intellectual disability with growth hormone deficiency	*SOX3*	15 → 26
Congenital central hypoventilation	*PHOX2B*	20 → 25–29
Promoter and 5′ untranslated region		
Fragile X syndrome	*FMR1*	5–50 → >200
Fragile X tremor/ataxia syndrome	*FMR1*	5–50 → 55–200
Fragile XE syndrome	*AFF2*	4–39 → 200–900
Progressive myoclonus epilepsy 1	*CSTB*	2–3 → 30–75
Spinocerebellar ataxia 12	*PPP2R2B*	4–32 → 43–78
Frontotemporal dementia and ALS	*C9orf72*	3–25 → >30
Neuronal intranuclear inclusion disease	*NOTCH2NLC*	7–60 → 61–500
Oculopharyngodistal myopathy 1	*LRP12*	13–45 → 90–130
Oculopharyngodistal myopathy 2	*GIPC1*	12–32 → 97–120
Oculopharyngeal myopathy with leukoencephalopa…	*NUTM2B-AS1*	3–16 → 40–60
Baratela-Scott syndrome	*XYLT1*	9–20 → 120–800
Developmental delay, ataxia, raised glutamine	*GLS*	8–16 → 680–1400
Introns		
Friedreich ataxia	*FXN*	5–34 → 66–1300
Spinocerebellar ataxia 10	*ATXN10*	10–32 → 280–4500
Myotonic dystrophy 2	*CNBP*	11–30 → 50–11000
Spinocerebellar ataxia 31	*BEAN1/TK2*	variable → 110–760
Spinocerebellar ataxia 36	*NOP56*	5–14 → 650–2500
Fuchs endothelial corneal dystrophy 3	*TCF4*	5–31 → >50
X-linked dystonia-parkinsonism	*TAF1*	none → 30–55
Spinocerebellar ataxia 37	*DAB1*	none → 31–75
Familial adult myoclonic epilepsy 1	*SAMD12*	variable → 440–3680
Familial adult myoclonic epilepsy 2	*STARD7*	9–20 → 661–735
Familial adult myoclonic epilepsy 3	*MARCHF6*	10–30 → 660–2800
Cerebellar ataxia, neuropathy, vestibular areflexia	*RFC1*	variable → 400–2000
3′ untranslated region		
Myotonic dystrophy 1	*DMPK*	5–37 → 50–10000
Spinocerebellar ataxia 8	*ATXN8/ATXN8OS*	15–50 → 74–250
Longer repeat units (25 and 99 bases)		
Myopathy with rimmed autophagic vacuoles	*PLIN4*	27–31 → 40
Alzheimer disease (risk factor)	*ABCA7*	12–427 → >200

A longer repeat at birth puts more neurons past 150 within a lifetime

The repeat a person inherits is the same in every cell at birth. From then on each cell's copy grows on its own, slowly at first and then faster once it passes about 90. A longer inherited length shortens the slow part, so more of that person's neurons reach 150 during a lifetime, and symptoms start earlier.

Born with 42 repeats

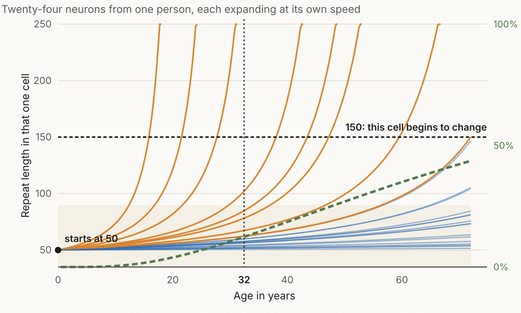
Twenty-four neurons from one person, each expanding at its own speed



Diagnosis is expected at 54. In this drawing 3 of the 24 cells pass 150 within a lifetime

Born with 50 repeats

Twenty-four neurons from one person, each expanding at its own speed



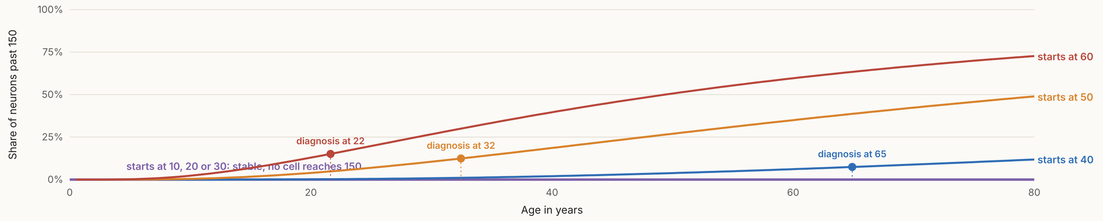
Diagnosis is expected at 32. In this drawing 8 of the 24 cells pass 150 within a lifetime

Dashed green line, right-hand scale: the share of that person's neurons that have passed 150 by that age

Shaded band: the slow phase, up to about 90 repeats. Each line is one neuron. The shapes and the spread follow the two-stage dynamic the two papers describe, with a growth exponent and a rate spread chosen to illustrate it. Expected age at diagnosis is 649/(CAG−30) years, from the CAG-Age Product as Enroll-HD standardizes it (CAP = 100 at the expected age of diagnosis; Warner et al. 2020).

Below about 36 repeats nothing happens; above it, every added repeat moves the whole curve left

Share of one person's neurons that have passed 150, by age, for six starting lengths



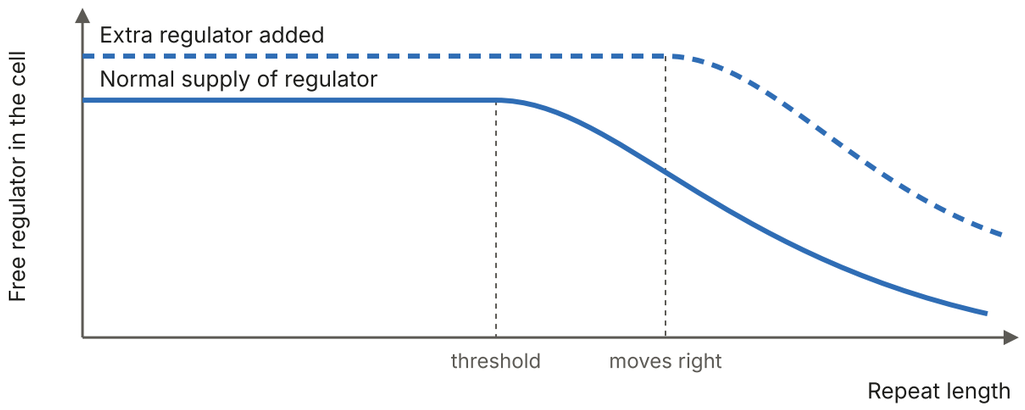
Dots mark the expected age at diagnosis, 649/(CAG−30) years. The curves follow from the drawn expansion speeds and are not fitted to onset data; that they put diagnosis at a similar share in all three cases is a consequence of the drawn model, not evidence for it. Lengths of 36 to 39 are shown here as stable, though in that range the disease sometimes appears late in life.

Source: Depienne and Mandel, American Journal of Human Genetics 108:764 (2021), Tables 1–3, lengths as reported to early 2021. Threshold of 150 CAGs: Handsaker et al., Cell 2025; Kashin et al., Cell 2026. No grey bar: normal length variable or absent. The fragile X tremor/ataxia row also covers premature ovarian insufficiency (same *FMR1* range). Hand-foot-genital syndrome: longest of three alanine tracts; infantile epileptic encephalopathy 1: first of two. Not drawn, for lack of reported length ranges: familial adult myoclonic epilepsy 4, 6 and 7, and risk-factor repeats in *CACNA1C* (schizophrenia, bipolar disorder) and *WDR7* (ALS).

What other repeat diseases suggest about the Huntington's disease threshold

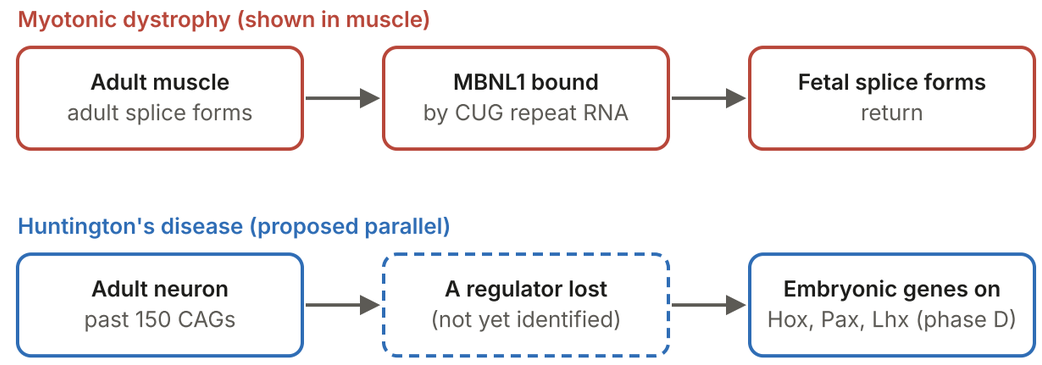
Inferences drawn from myotonic dystrophy, fragile X and facioscapulohumeral dystrophy. None has yet been tested in Huntington's disease.

1. A sharp threshold can come from a repeat binding up a limited supply of a regulator



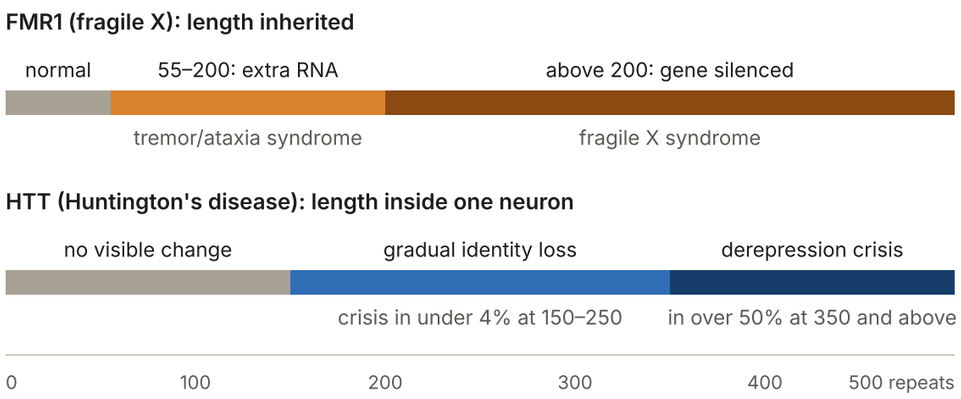
In myotonic dystrophy the CUG repeat RNA binds MBNL1 until the cell's supply runs short, and raising MBNL1 rescued model mice. Expanded polyglutamine binds the co-activator CBP, and raising CBP rescued cultured neurons. For Huntington's disease, raising the bound regulator should move the threshold above 150 CAGs.

2. Losing a regulator can return a cell to an earlier developmental program



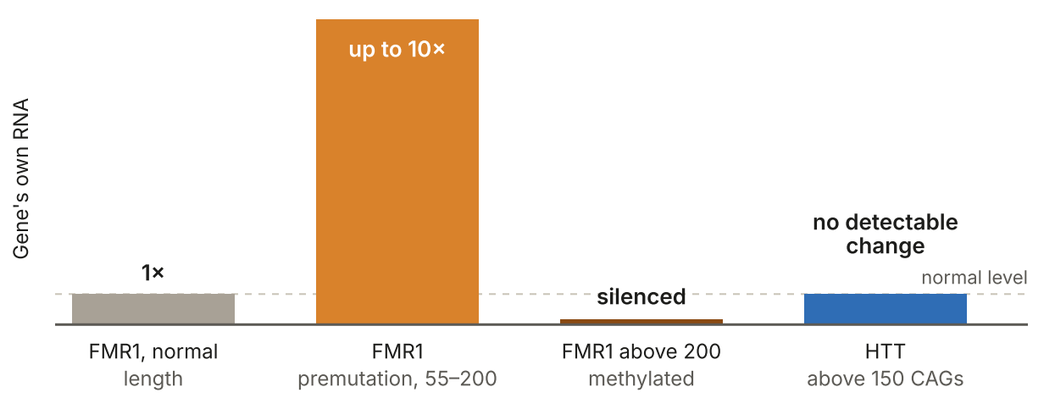
Without MBNL1, adult muscle keeps fetal splice forms. In phase D, Huntington's disease neurons switch on embryonic patterning genes. PRC2, which keeps those genes silent, is the obvious candidate regulator; the direct evidence so far shows mutant huntingtin stimulating PRC2 rather than inhibiting it.

3. One repeat can have two thresholds, each with its own mechanism



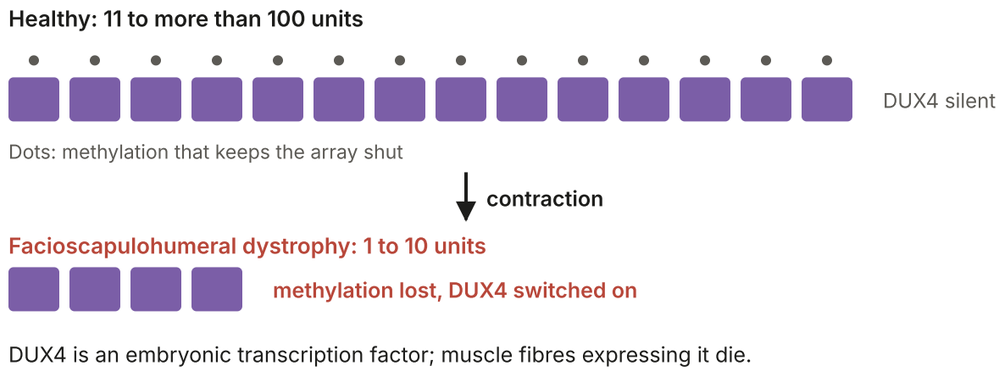
The *FMR1* repeat produces extra toxic RNA at 55–200 and silences its gene above 200. Huntington's disease neurons lose identity genes gradually from 150 CAGs, while the derepression crisis becomes common only above about 350 and does not grow with length once begun, which fits a second mechanism with its own threshold.

4. Huntington's disease differs from the silencing diseases



Long *FMR1* repeats raise or silence their own gene's RNA. The rise runs with repeat length across the premutation range and reaches ten times normal at the long end. *HTT* RNA showed no detectable change past 150 CAGs, so silencing of *HTT* itself does not explain the threshold. The sequencing used could not see how *HTT* RNA is processed, which points to measuring the short HTT1a fragment directly.

5. One disease is caused by the repeat shortening

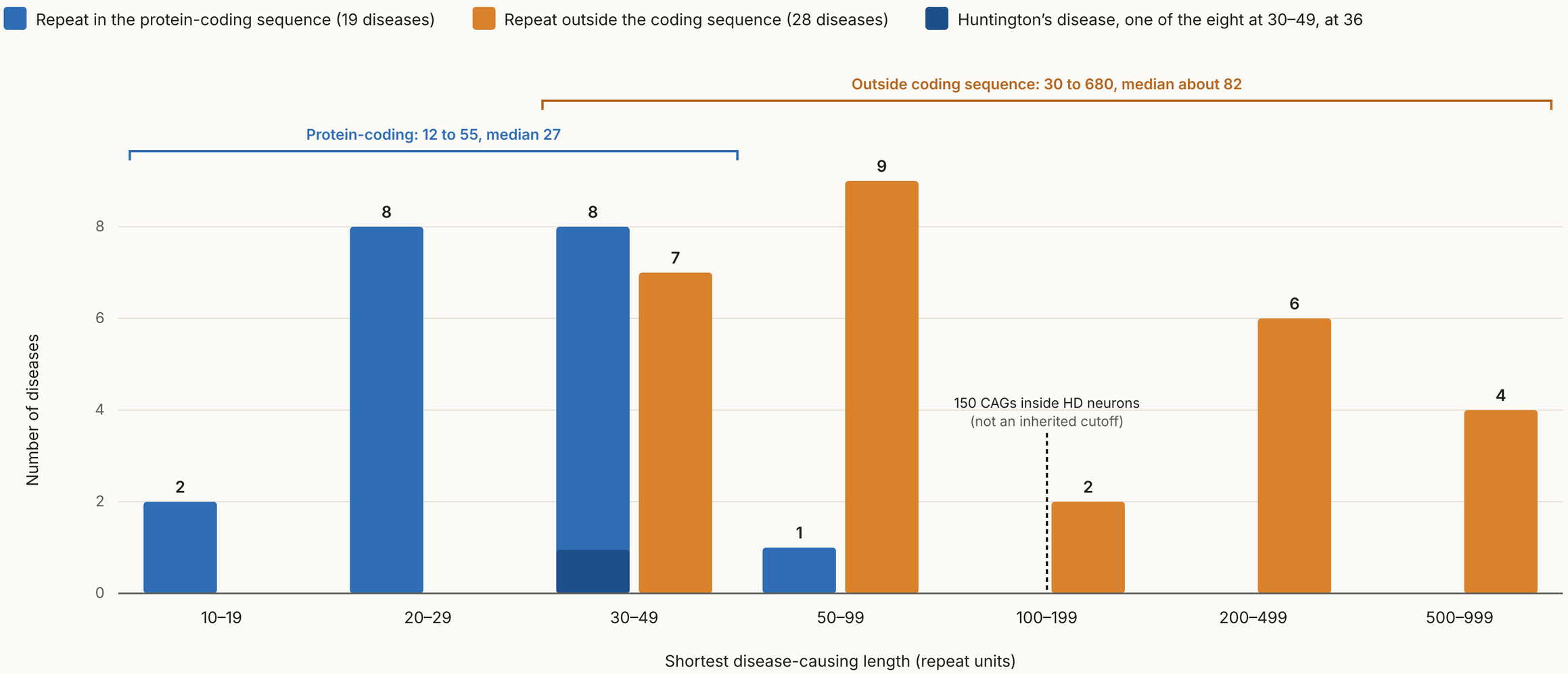


Shortening the D4Z4 array below about 11 units strips the methylation that silences it and switches on *DUX4*, a gene used only in the early embryo. The endpoint matches the derepression crisis in Huntington's neurons, where developmental transcription factors come on in a cell that should have them shut, but it is reached by the repeat getting shorter rather than longer. A short array causes disease only on a chromosome carrying the 4qA variant, and a second form arises at normal length when *SMCHD1*, which maintains the methylation, is mutated: what matters is the silencing, not the length by itself.

Sources: Lin et al., Hum Mol Genet 2006; Kanadia et al., PNAS 2006; Nucifora et al., Science 2001; Seong et al., Hum Mol Genet 2010; Tassone et al., Am J Hum Genet 2000; Depienne and Mandel, Am J Hum Genet 2021; Handsaker et al., Cell 2025; Kashin et al., Cell 2026; Cascella et al., Front Neurol 2018.

Shortest disease-causing repeat length across 47 repeat diseases

Each disease counted once, at the lowest length in its disease-causing range, in repeat units. Bins widen as lengths grow. Cutoffs reported as “above” a number are counted at that number.



1 How the repeat does damage sets the cutoff

Repeats in the coding sequence act through the protein and cause disease at 12–55 units. Repeats elsewhere act through toxic RNA, gene silencing or repeat proteins and need 30–680. CTG causes disease at 50 in *DMPK* and 74 in *ATXN8OS*; CAG causes Huntington's disease at 36 inside the coding sequence.

2 Gene-silencing diseases sit near 200

Fragile X (above 200), fragile XE (200) and Baratela-Scott syndrome (120) are CGG or CCG repeats in the region that controls the gene. The expanded repeat becomes methylated and the gene shuts down, which fits one mechanism with its own length requirement.

3 The glutamine cluster has two explanations

Ten CAG diseases in ten genes start at 32–55. Polyglutamine begins to aggregate near that length; the 2025 Cell paper reads 36–40 in Huntington's disease as the length at which the DNA repeat becomes unstable enough to reach 150 in a lifetime. Both depend on CAG length, so the histogram does not separate them.

4 Proteins respond to a few extra units

Alanine tracts cause disease with 4–10 extra residues (*RUNX2* 17 to 27, *PABPN1* 6–10 to 12); their repeats are stable and the diseases mostly congenital. Huntington's disease neurons show no detectable change from 36 to 150 CAGs, which fits toxicity arising at long lengths by another route.

5 SCA6 works without somatic expansion

SCA6 causes disease at 20–33 CAGs, and its repeat shows no mosaicism across brain regions or in sperm; a review rates its instability very low or absent, against high for SCA1, 2, 3 and 7. The glutamine diseases therefore include at least one that does not depend on expansion to a high threshold.

6 The long end reflects discovery history

The split between 30–120 and 200+ outside the coding sequence may shift as more repeats are found. Very long repeats were hard to detect before long-read sequencing, and most cutoffs of 200–680 belong to intronic repeats found with it since 2017.

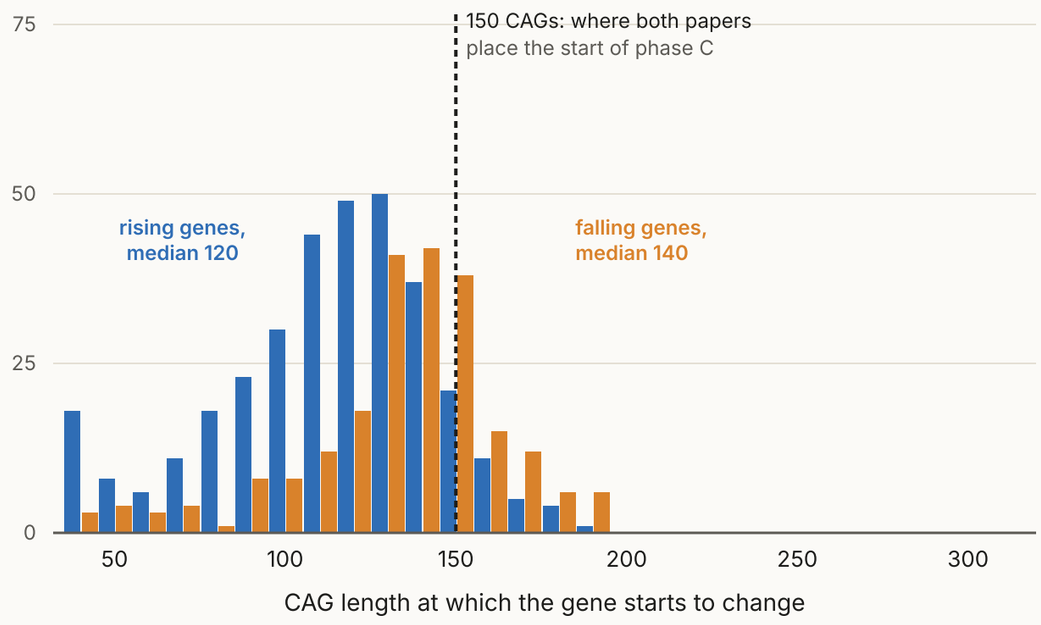
Sources: Depienne and Mandel, Am J Hum Genet 2021 (lengths); Handsaker et al., Cell 2025; Mantuano et al., Cytogenet Genome Res 2003, citing Ishikawa et al. 1999 (brain) and Shizuka et al. 1998 (sperm) for SCA6 stability. Inferences 1–4 and 6 are drawn from the histogram and have not been tested.

Re-analysis of the data released with the two HD papers

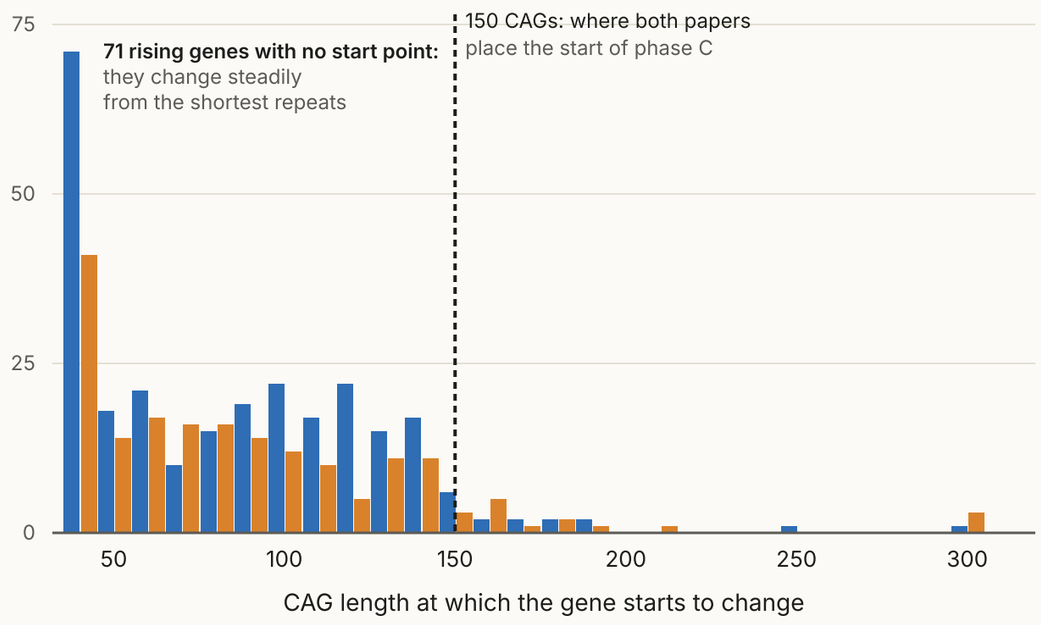
Nuclei with a measured repeat above 35 CAGs from the 2026 cortex paper's archive (Zenodo record 18202664). Phase C genes as the authors defined them ($p < 10^{-10}$). Striatal projection neurons were grouped by donor only, so D1-type and D2-type neurons are pooled; cortical neurons were grouped by subtype.

Gene rises as the repeat grows Gene falls as the repeat grows

Striatum: 557 genes, most start between 100 and 170

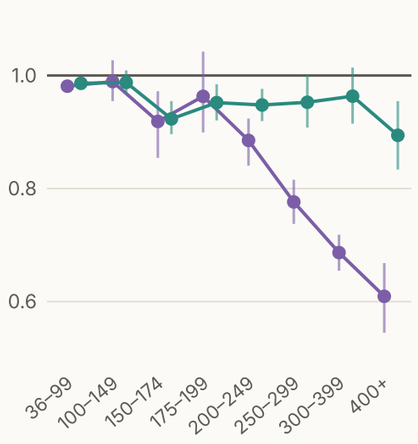


Cortex: 446 genes, half change from the shortest repeats

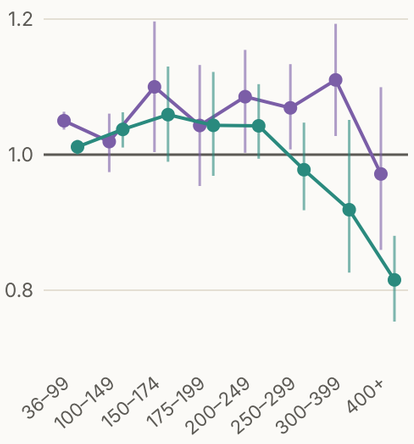


Striatum Cortex Expression relative to nuclei with 36–99 CAGs (1 = no change), by CAG length; lines show 95% intervals

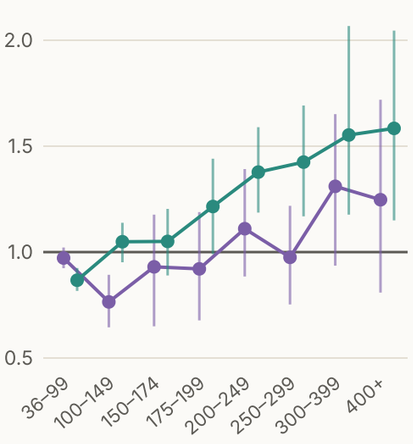
MBNL2
Falls in the striatum from 150 CAGs



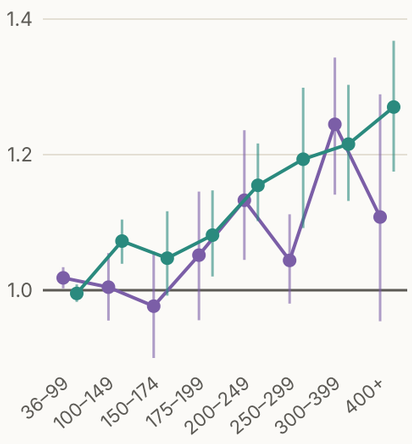
HTT
Falls in the cortex past 250 CAGs



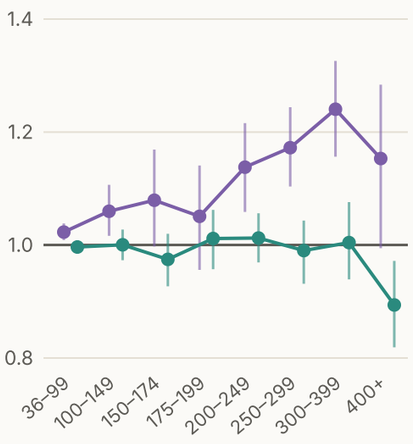
EZH2
PRC2; rises in the cortex



CREBBP
CBP; rises in the cortex



TCERG1
Rises in the striatum only



Striatal nuclei past 150 CAGs: those with an inclusion are further along

With an inclusion (159 nuclei)



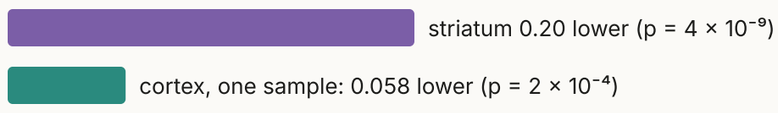
60% in the late, failing phase

Without an inclusion (20 nuclei)



25% in the late, failing phase

Falling identity genes, score difference

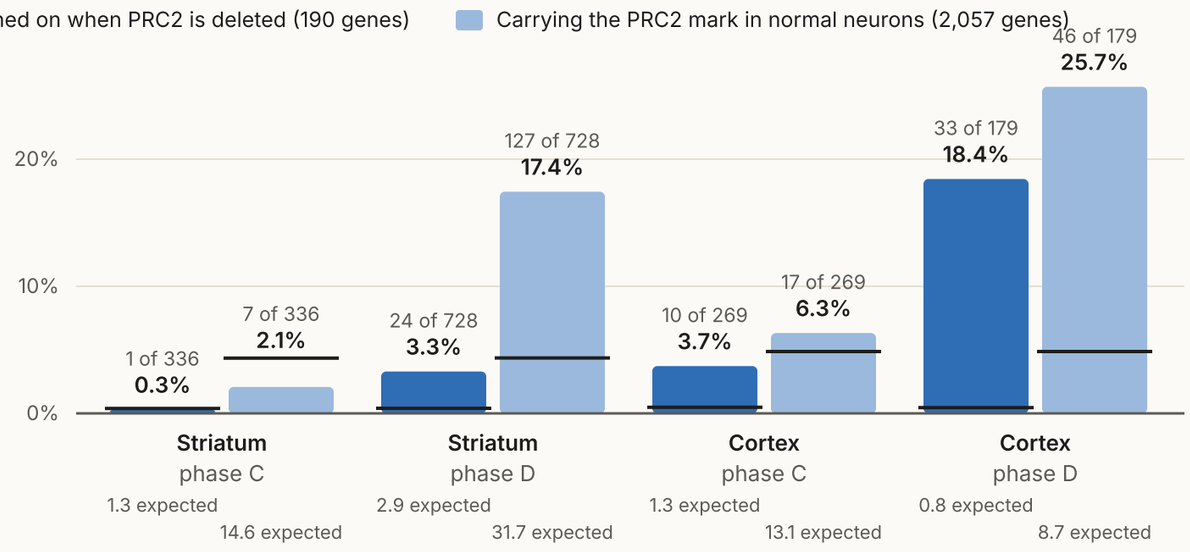


Each circle stands for 5% of nuclei; orange = late, failing phase; dark dot = inclusion.
Repeat length was not matched between the two sets, and nuclei with inclusions have much longer repeats, so this is consistent with the authors' account and adds nothing to it.

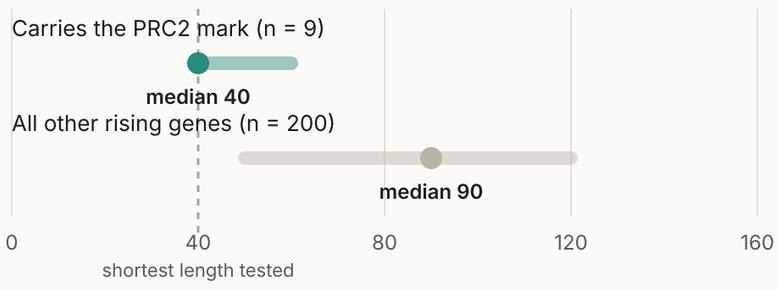
Phase D genes are largely the genes a neuron switches on when PRC2 is removed

The 2026 cortex paper raises a loss of polycomb function in phase D as a possibility and leaves it untested.
Bars show the share of each gene set that appears in two gene lists from von Schimmelmann et al. 2016.
Tick marks give the share expected by chance, which differs between the lists.

Switched on when PRC2 is deleted (190 genes) Carrying the PRC2 mark in normal neurons (2,057 genes)



Cortical rising genes, phase D genes removed
CAG length at which the gene starts to change, middle half and median

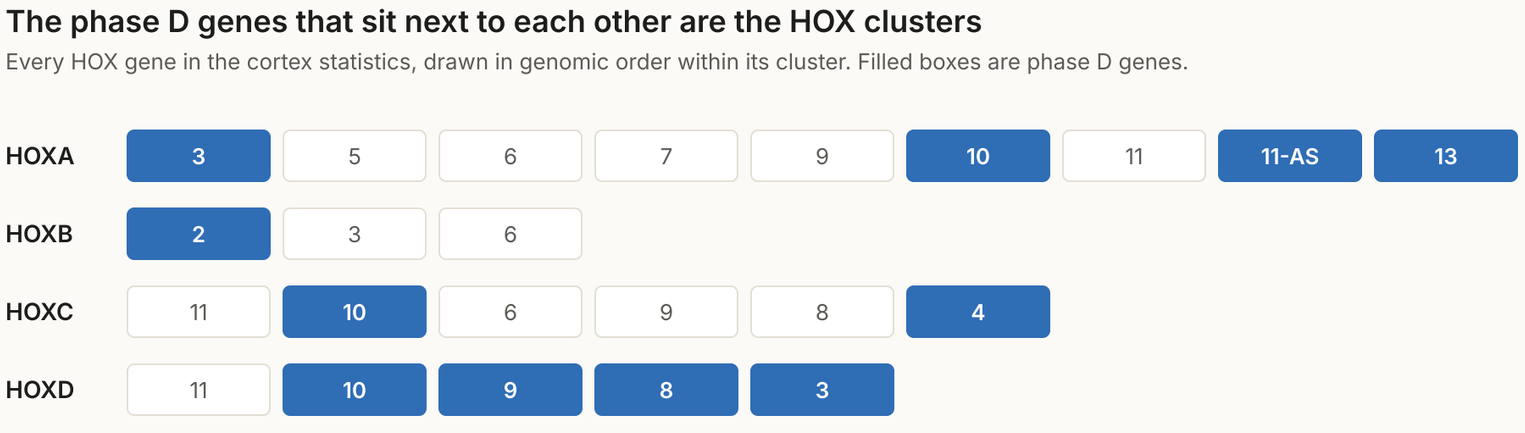


Six of the nine marked genes start at 40, the shortest length in the fit, so their onset could lie lower. p = 0.03, and not significant after correcting for the four tests run.

Seven of the ten cortical phase C genes on the knockout list are also phase D genes. With every phase D gene removed from both sides, three remain, a fourfold enrichment at p = 0.04.
The striatum shows no phase C enrichment. The reference gene lists are from mouse, matched to the human genes by name, and the silencing map was measured in striatal neurons.

The phase D genes that sit next to each other are the HOX clusters

Every HOX gene in the cortex statistics, drawn in genomic order within its cluster. Filled boxes are phase D genes.



Cortex

11 of the 23 HOX genes tested are phase D genes.
Chance would put 0.22 of them there.

Striatum

4 of 17, against 0.54 expected.

What else runs together

MEG3 with MEG8 on chromosome 14, the imprinted DLK1-DIO3 locus, whose silencing is held by DNA methylation at a control region. Nothing else.

Set the clusters aside and the remaining phase D genes are no more bunched than chance, so what they share is being a polycomb target rather than being neighbours. Polycomb targets lie all over the genome; HOX is where they sit in a row.

The clusters bring their own long non-coding RNAs with them — HOTAIRM1, HOTTIP, HOXA-AS2, HOXD-AS2 and HOXB-AS1 all reach phase D — and the HOX genes rise from below 1 transcript per million to a level still below it, which is what a silenced domain opening looks like rather than a gene switched on hard.

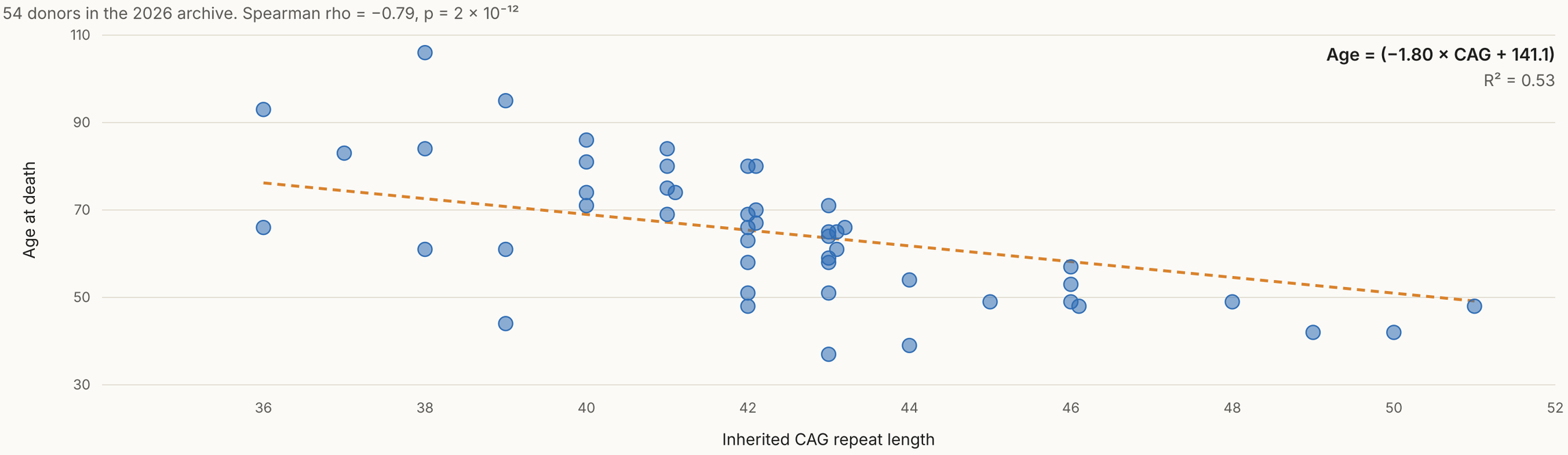
Counted by gene family. A spatial test is not used for these genes: the phase sets run near 190 transcripts per million against 17 for all tested genes, so a null drawn from every gene is mis-matched, and matching on abundance drops all 11 HOX genes, which sit below the 1 TPM floor of the abundance table.

How fast the repeat grows, and what slowing it would buy

Donor records released with the 2026 cortex paper (Zenodo record 18202664): age at death, the repeat length inherited on the affected copy of *HTT*, and per-nucleus repeat lengths in cortical neurons for 11 of the donors. Expected onset ages are 649/(CAG−30) years, the Enroll-HD calibration of the CAG-Age Product.

Age at death falls as the inherited repeat lengthens

54 donors in the 2026 archive. Spearman rho = −0.79, p = 2×10^{-12}

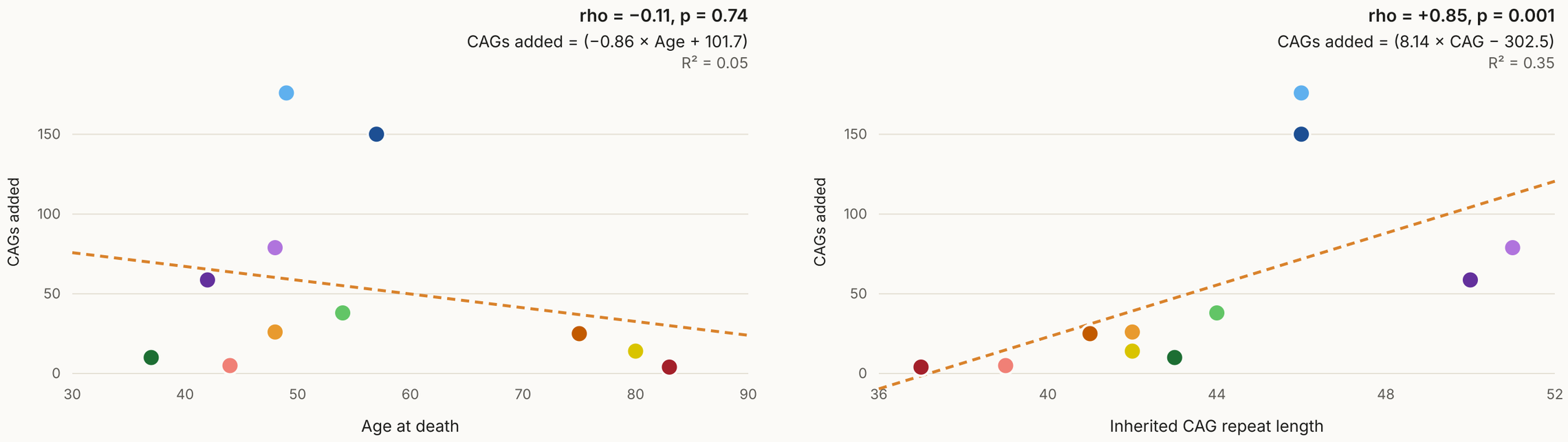


Age at death in this group is set by the disease, so the donors who had the most years are the ones whose repeats grow slowest.

A longer stretch of time and a faster expansion never occur together, and that pairing is the comparison which would give the growth rate a number.

In the 11 donors with per-nucleus lengths, the inherited length predicts the expansion and the age does not

Expansion is the 90th percentile repeat length in that donor's cortical neurons, less the length inherited



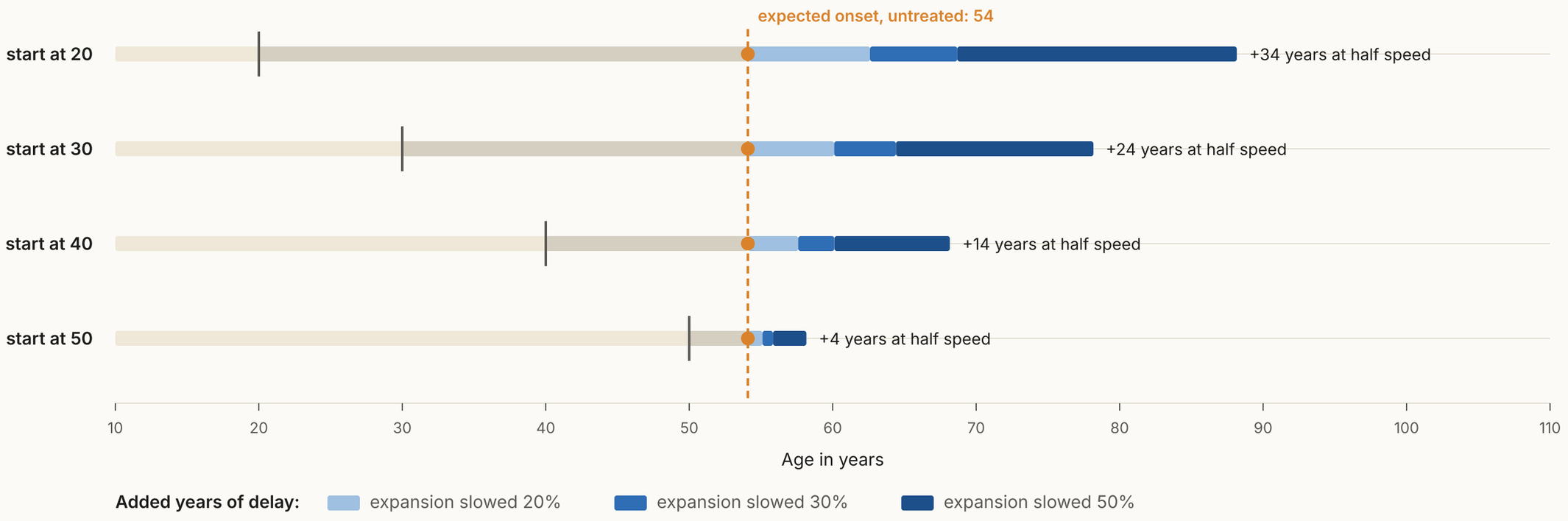
The flat panel is what blocks the measurement. Time leaves no mark on how far the repeat has travelled, so the two quantities a growth rate needs, distance and duration, cannot be separated in this material.

Each donor keeps its colour on both panels, the colours running in spectral order by inherited length.

The left line falls by less than one CAG per year of age and accounts for 5% of the difference between donors; the right line accounts for 35%.

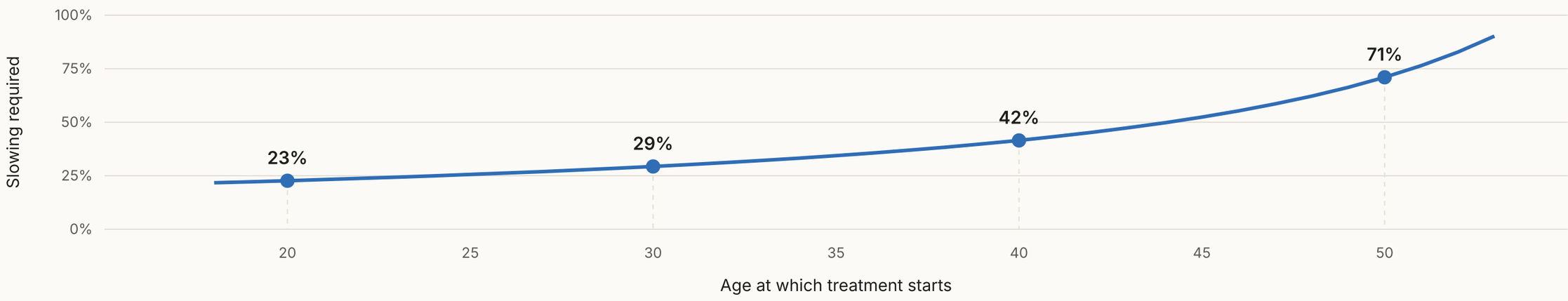
Slowing expansion stretches the time that is left, so an early start is worth more than a strong effect

A person born with 42 repeats, expected onset at age 54. Rate and time enter any growth law only as their product, so cutting the rate by a fraction f multiplies every remaining interval by 1/(1−f).



The same relationship read backwards: how much slowing ten years of delay would take

Born with 42 repeats, expected onset at age 54



Silencing MSH3 by about 60% blocked measurable expansion in the striatum of BAC-CAG mice over the four months followed. A partial knockdown produced a near-complete stop there, so the percentages on this curve are slowing of the expansion itself, not a dose of knockdown.

Read against the curve: ten years of delay takes 29% slowing if it begins at age 30, 42% at 40 and 71% at 50.

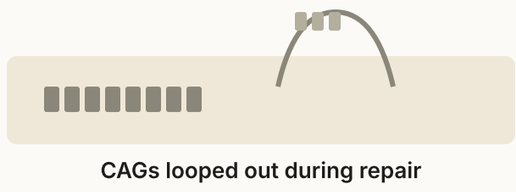
Sources: donor records and per-nucleus repeat lengths, Kashin, Lee et al., *Cell* 2026 (Zenodo record 18202664). Expected onset from the CAG-Age Product, Warner et al. 2020. MSH3 silencing in mice: O'Reilly et al., *Molecular Therapy* 31:1661, 2023; the four-month figures are read from the bioRxiv preprint of that paper. The delay and slowing curves are arithmetic on the onset formula, not fitted to data.

FAN1, the gene that ranks first across the 11 cortex donors

A re-analysis of the single-nucleus data released with the 2026 cortex paper (Zenodo record 18202664). Every gene in the matrix was given a per-donor expression rate standardized on the distribution of repeat lengths, then correlated with how far each donor's repeats had travelled. The gene that comes out first is *FAN1*, the nuclease that removes the looped-out repeat.

Two enzymes act on the same intermediate, and they pull in opposite directions

A repeat tract being copied can slip, leaving a stretch of CAGs looped out of the duplex. What happens to that loop decides whether the repeat ends longer or unchanged.



MutSβ nicks the opposite strand

MSH2 and MSH3 together, with MutLγ. The gap is filled using the loop as the template, so the extra CAGs are written into the sequence.

11 triplets

The repeat ends longer than it started.

FAN1 cuts the looped strand

A structure-specific nuclease, acting with PCNA and RFC. The loop is removed and the strand resynthesized, so the extra CAGs never enter the sequence.

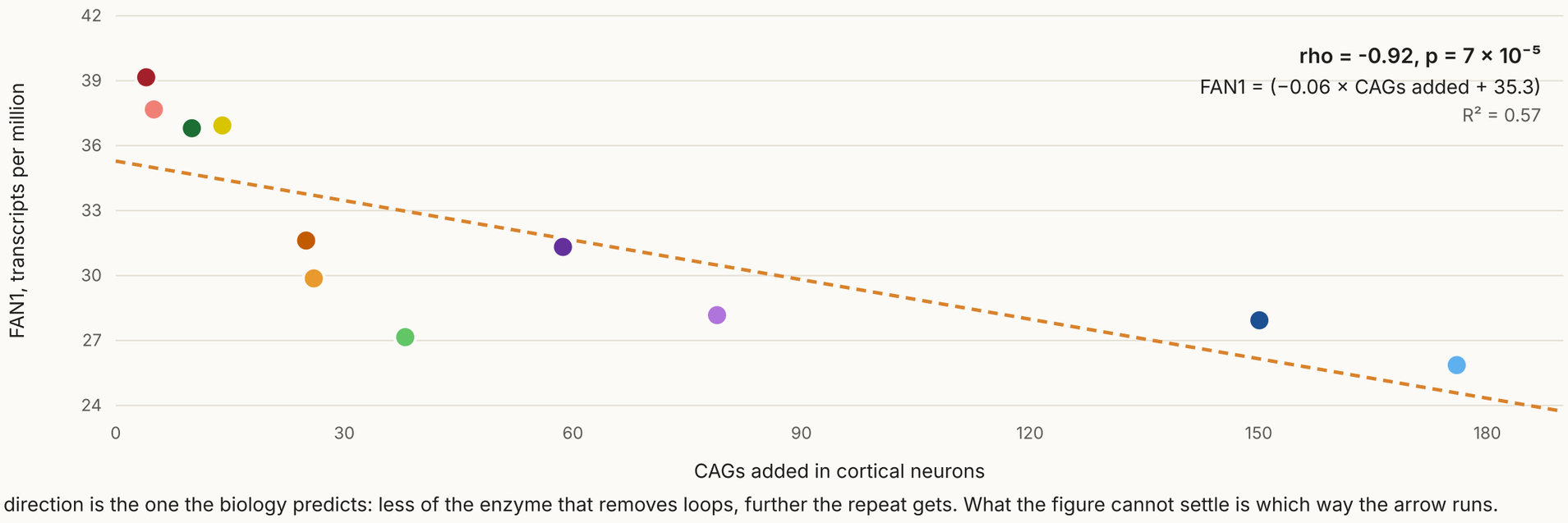
8 triplets

The repeat returns to its original length.

The two are not independent: FAN1 also inhibits MutLγ, the nuclease MutSβ needs to make the nick, so it restrains the expansion step as well as removing the loop. The direction a cell drifts is set by the balance between them, not by either alone.

Across the 11 cortex donors, the ones whose repeats travelled furthest carry the least FAN1

FAN1 transcripts per million in cortical excitatory neurons, standardized so every donor is compared at the same distribution of repeat lengths. Expansion is the 90th percentile repeat length in that donor, less the length inherited.



Holding the inherited length constant

rho = −0.82

p = 0.004

Longer inherited repeats both expand faster and sit lower in FAN1, so the partial figure matters.

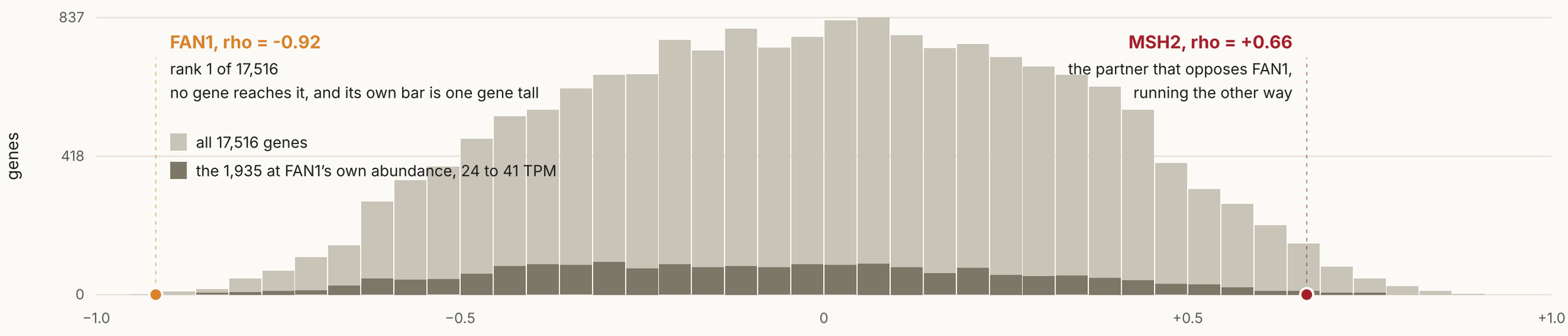
It is the rank correlation within each inherited length, pooled.

The direction is the one the biology predicts: less of the enzyme that removes loops, further the repeat gets. What the figure cannot settle is which way the arrow runs.

If the growth of the repeat, or the damage that follows it, lowers FAN1 in the neurons furthest along, the same pattern appears with the arrow running the other way. Only material sampled before the expansion could tell the two apart.

Every gene in the matrix was standardized; no gene tracks expansion more closely than FAN1

17,516 genes reaching 1 transcript per million, each correlated with expansion across the same 11 donors. Expansion orders the donors, so any gene varying along that order produces a correlation: this is the distribution of what that alone yields.



The darker part of each bar is the 1,935 genes expressed within a factor of 1.3 of FAN1 itself. They are spread as widely as the rest, a standard deviation of 0.33 against 0.34, and the nearest of them reaches 0.85.

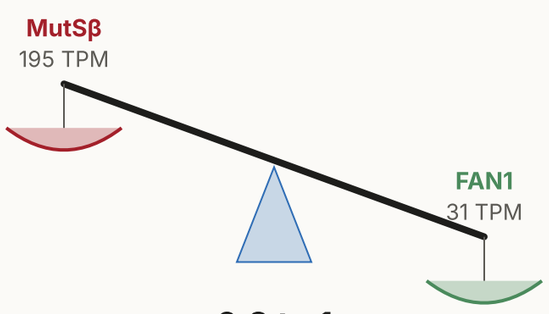
That is a reference point, not a p-value: genes are correlated with one another, so the tail is not 17,516 independent draws.

The direction MSH2 runs is the one the mechanism implies, from the opposite side of the balance.

The striatum, where the repeats run furthest, is where FAN1 is most outnumbered — and where 6 donors cannot settle anything

Regional pseudobulk expression from the same matrices, and the same standardized correlation run on striatal projection neurons.

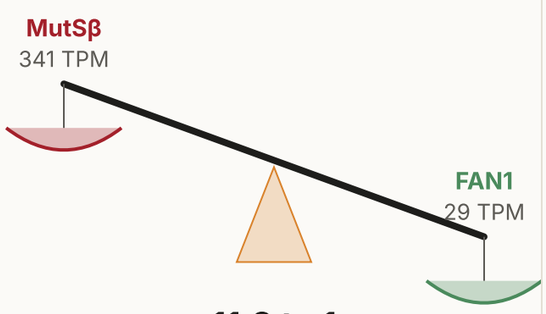
Cortical excitatory neurons



6.2 to 1

median 26 CAGs added

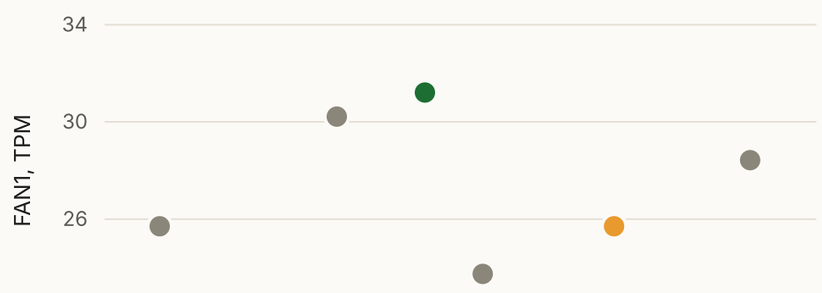
Striatal projection neurons



11.8 to 1

median 145 CAGs added

Striatum: 6 donors, no relationship rho = -0.03



Two of the six are also cortex donors and keep their colours; the other four are grey.

The two regions do not disagree. The striatal donors all sit far out along the expansion axis, between 72 and 205 CAGs added, and their FAN1 values span barely 8 transcripts per million.

With 6 donors and almost no spread in either quantity, a correlation near zero is what a real effect of the cortical size would most often produce here. The striatum is not evidence against; it is close to no evidence either way.

Taken together

The cortex result is one cohort of 11 and an association, and the direction of cause is open. It is worth a test in material where repeat length is known before the neuron is lost: FAN1 measured in presymptomatic tissue, or in carriers matched on inherited length and differing in onset. The mechanism already has independent support and the modifier genetics already point at FAN1; what is new here is that the ordering holds between individuals, within one tissue.

The standardization, the barcode matching and the streaming reader of the expression matrices are at github.com/crgfrench/hd-somatic-expansion; the figures above come from that code.

Sources: single-nucleus expression and per-nucleus repeat lengths, Kashin, Lee et al., Cell 2026 (Zenodo record 18202664), cortical excitatory neurons and striatal projection neurons. Repair mechanism: Senoussi et al., Nature Communications 16:9445, 2025.